# Hari Varshan

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## SUMMARY

Computer Science Ph.D. researcher developing foundation and generative models for medical imaging and physiological time series. Experience with multimodal learning, 3D imaging, wearable data, and distributed multi-GPU training across clinical and public datasets.

## EDUCATION

**Ph.D. in Computer Science** (M.S. in Computer Science awarded en route) **August 2025 – Present**
- Advised by Dr. Teresa Wu GPA: 4.00/4.00
- Research Areas: Foundation Models, Generative AI, Multimodal Learning, Medical Imaging, Physiological Time Series
- Applications: Precision Health, Early Prediction & Detection, Computational Oncology, Neuroimaging, Wearable Health

Arizona State University, Tempe, AZ

**Bachelor of Science in Computer Science** **May 2025**
- Honors: Barrett, The Honors College

**Bachelor of Science in Mathematics** **May 2025**
- Honors: Barrett, The Honors College

Arizona State University, Tempe, AZ

## RESEARCH & PROFESSIONAL EXPERIENCE

**Wu Lab, ASU–Mayo Center for Innovative Imaging, Tempe, AZ: Graduate Research Associate** **August 2025 – Present**
*Advisor: Teresa Wu, Ph.D.*
- Build end-to-end deep learning pipelines for 3D medical image segmentation, anomaly detection, and multimodal classification across 10,000+ 3D volumes from 9 public and clinical datasets.
- Adapt and evaluate foundation models across medical imaging (DINOv2, BioMedCLIP) and physiological time series (Moirai, MOMENT), using multi-GPU distributed training on SLURM with reproducible experiment tracking and evaluation.
- Develop wearable time-series pipelines using minute-level Fitbit data from 1,158 All of Us participants, applying foundation-model representations to longitudinal physiological signals for migraine phenotype classification.

**Complex Adaptive Systems Initiatives (CASI), Scottsdale, AZ: AI Research Intern** **May 2025 – August 2025**
*Advisor: Chief Scientist George Poste, Ph.D., DVM*
- Led the research design of a multimodal deep learning framework for lung adenocarcinoma prognosis, integrating whole-slide histopathology, WES/WTS genomic data, and clinical records to predict disease progression and treatment response.
- Designed a multimodal architecture combining CNN + Multiple Instance Learning (MIL) for whole-slide pathology with MLP-based genomic encoders and intermediate feature fusion for joint prognostic modeling.
- Led 2 graduate and 1 undergraduate researcher and developed the 12-month experimental roadmap for a proposed **ASU–Caris** Life Sciences collaboration, spanning data curation, baselines, validation, and interpretability.

**Meteor Studio Lab, ASU, Tempe, AZ: Student Researcher** **August 2024 – May 2025**
*Advisor: Robert LiKamWa, Ph.D.*
- Co-led the development and implementation of a pre-op inspection simulator for Komatsu using Unreal Engine and HaptX Gloves, saving $50K/hour by reducing training time on actual mining trucks.
- Engineered production VR/AR software using Unity, Unreal, and C# across 5+ concurrent client projects (Scottsdale Water, Kiewit Construction, ASU), shipping on Quest 2/3, HTC Vive, and PSVR at a sustained 90+ FPS, while effectively managing a team of 6 software developers using Git and Plastic SCM as our main workflow tools.
- Developed an XR development framework and reusable Unity package for automated timeline generation and interaction systems, currently integrated into a college course that teaches 100+ students each semester to rapidly prototype timeline-based XR productions.

**DocYou LLC, Tempe, AZ: Co-Founder** **August 2022 – May 2025**
- Co-founded and developed a dynamic web and mobile application utilizing React Native, PHP, and Firebase, and Figma for Phoenix Children's Hospital, streamlining medical procedure documentation and ensuring ACGME accreditation for 50+ residents.
- Integrated visualization tools using JavaScript's D3 and Plotly, providing residential supervisors with comprehensive insights into procedure logging for an average of 20 residents, enhancing efficiency and accountability.
- Championed iterative development with Agile methodology and Sprint framework, adeptly integrating project management, Git commands, and DevOps practices, resulting in the acquisition of $16,000 in funding through pitch competitions and tech conferences.

**DATAD. CO, Richardson, TX: Data Science Intern** **May 2024 – July 2024**
- Developed and prompt-engineered an LLM-based conversational system using EleutherAI models and the OpenAI API for domain-specific information retrieval and interaction.
- Implemented ML models using time-series data to predict scope of incident probability and safety risk for underground miners.
- Developed client-facing safety-compliance applications and KPI dashboards using Power Apps, Power Automate, Power BI, and Logic Apps to support operational and financial decision-making.

## RESEARCH PROJECTS

**Unpaired Generative Learning for Amyloid PET Harmonization** *Research Project | Manuscript in Preparation*

- Developed a Feature-Mapping CycleGAN (FM-CycleGAN) to harmonize regional amyloid PET measurements across PiB and FBP tracers without requiring paired scans for model training.
- Trained and optimized the framework on 5,433 unpaired PET scans across multiple neuroimaging cohorts using residual MLP generators and adversarial, cycle-consistency, and identity losses.
- Improved inter-tracer mcSUVR correlation from **R = 0.922 to R = 0.963** on an independent paired test cohort ($p < 0.001$), demonstrating effective PET harmonization without paired training data.

**Multimodal Deep Learning for Incidental Thyroid Nodule Risk Stratification** *Research Project | Manuscript in Preparation*

- Developed 2D, 3D, and multimodal deep learning models to classify benign vs. malignant incidental thyroid nodules from CT images and radiology-report text across a 454-patient clinical cohort.
- Fine-tuned **BioMedCLIP** vision and text encoders and benchmarked against 2D ResNet/U-Net and MONAI 3D architectures using patient-level 5-fold cross-validation and held-out evaluation.
- Achieved **0.768 AUC** with multimodal image-text fusion on a 91-patient internal holdout cohort, outperforming image-only and volumetric model variants.

**Foundation Models for Cognitive Impairment Detection from Naturalistic Driving** *Ongoing Research Project*

- Developed subject-separated time-series pipelines for MCI detection from naturalistic driving, processing accelerometer and gyroscope signals from 19 older adults into trip-level windows and high-frequency 24 Hz turn segments.
- Identified substantial driver-specific structure in foundation-model embeddings and designed leave-pair-out evaluation with permutation testing to prevent subject leakage and inflated MCI classification performance.
- Improved frozen MOMENT performance from **0.51 AUROC** on trip-level windows to **0.64** on turn-level dynamics and **0.79** on right turns, indicating that high-frequency maneuver-level representations preserve candidate MCI-related signal lost through coarse trip-level aggregation.

**Wearable Time-Series Foundation Models for Migraine Subtyping** *Ongoing Research Project*

- Developed a reproducible wearable time-series pipeline using minute-level Fitbit heart-rate data from 1,158 All of Us participants for chronic vs. episodic migraine classification.
- Processed longitudinal physiological signals into fixed-length sequences by detecting device artifacts, resampling irregular observations to minute-level grids, and constructing 20-day modeling windows.
- Evaluated time-series foundation-model embeddings against statistical baselines using subject-level 5-fold cross-validation to characterize the separability of chronic and episodic migraine phenotypes.

## TECHNICAL SKILLS

**ML/AI:** PyTorch, DINOv2, CLIP/BioMedCLIP, Moirai, MOMENT, ViTs, GANs/CycleGAN, diffusion models, self-supervised learning, multimodal learning, MONAI, MedSAM, scikit-learn
**Training & Systems:** PyTorch DDP, CUDA, SLURM/HPC, Docker, Kubernetes, Azure, Weights & Biases, TensorBoard, Git/GitHub
**Programming:** Python, R, C/C++, MATLAB, SQL, JavaScript, Bash
**Medical/Scientific:** NiBabel, SimpleITK, FreeSurfer, NumPy, Pandas, SciPy
**Additional:** FastAPI, PostgreSQL, MongoDB, React, Node.js, Django

## HONORS & SERVICE

- **Reviewer** | *Journal of Alzheimer's Disease* — July 2026 – Present
- **University Graduate Fellowship** | *Arizona State University* — February 2026
- **Graduate Teaching Associate** | *Arizona State University* — August 2025 - December 2025
- **School of Computing and Augmented Intelligence Student Mentor** | *Arizona State University* — May 2022 – May 2024